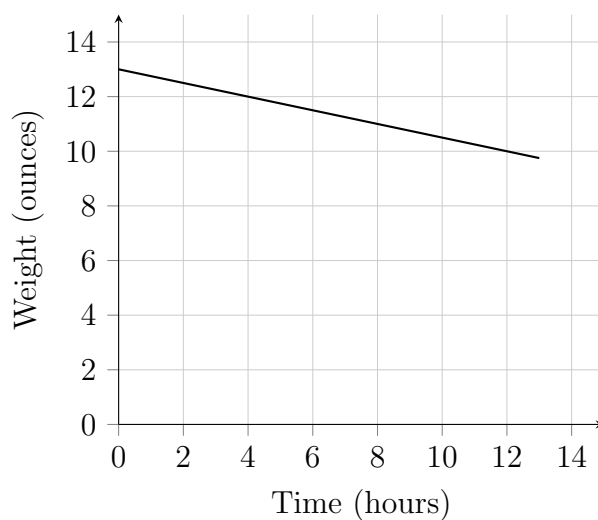


2026 March Int II

Module 1

Question 1



The function w models the weight, in ounces, of the remaining solid wax of a burning candle as a function of the time x , in hours, after the candle was lit. The graph of $y = w(x)$ is shown. According to the model, what was the weight, in ounces, of the candle's remaining solid wax 12 hours after the candle was lit?

- A) 4
- B) 8
- C) 10
- D) 13

Question 2

In $\triangle ABC$, the sum of the measures of $\angle A$ and $\angle B$ is 159.5° . What is the measure of $\angle C$?

- A) 20.5°
- B) 90°
- C) 110.5°
- D) 180°

Question 3

A local gym is offering a 25% discount on an annual membership. The regular cost of an annual membership is \$288. How much less is the discounted cost of the membership than the regular cost of the membership?

- A) \$25

- B) \$72
- C) \$216
- D) \$288

Question 4

x	y
0	8
1	9
2	10

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . Which of the following equations represents this relationship?

- A) $y = 8x$
- B) $y = 10x + 2$
- C) $y = 10x$
- D) $y = x + 8$

Question 5

The function f is defined by the equation $f(x) = 380x$. What is the value of $f(10)$?

Question 6

An object traveled 0.20 meters for every 4.0 seconds it was in motion. If the object was in motion for 24 seconds, what is the total distance, in meters, the object traveled?

Question 7

$$\begin{aligned}y &= 0 \\y &= x^2 - 36\end{aligned}$$

Which ordered pair (x, y) is a solution to the given system of equations?

- A) (0, 6)
- B) (0, 36)
- C) (6, 0)
- D) (36, 0)

Question 8

When a scientist began observing a sample of an isotope, the mass of the sample was 530,000 milligrams. The mass of the sample decreases by half approximately every 5 days. Which of the following is closest to the mass, in milligrams, of the sample 35 days after the scientist began observing the sample?

- A) 4,141
- B) 15,143
- C) 16,563
- D) 75,714

Question 9

The height of a right circular cylinder is 31 inches, and the circumference of its base is 310 inches. Which expression represents the total surface area, in square inches, of the cylinder?

- A) 31310
- B) $\pi\left(\frac{155^2}{\pi}\right)31$
- C) $31310 + \pi\left(\frac{155^2}{\pi}\right)$
- D) $1310 + 2\pi\left(\frac{155^2}{\pi}\right)$

Question 10

The function $f(w) = 6w^2$ gives the area of a rectangle, in square feet (ft^2), if its width is w ft and its length is 6 times its width. Which of the following is the best interpretation of $f(13) = 1,014$?

- A) If the width of the rectangle is 13 ft, then the area of the rectangle is 1,014 ft^2 .
- B) If the width of the rectangle is 13 ft, then the length of the rectangle is 1,014 ft.
- C) If the width of the rectangle is 1,014 ft, then the length of the rectangle is 13 ft.
- D) If the width of the rectangle is 1,014 ft, then the area of the rectangle is 13 ft^2 .

Question 11

The expression $(3x + 5)(8x - 7)$ can be written in the form $ax^2 + bx + c$, where a , b , and c are constants. What is the value of $a + b$?

Question 12

The ratio a to b is equivalent to the ratio 49 to 26. What is the value of a when $b = 182$?

Question 13

At the beginning of the year, Imani has \$170 in her account. Each month this year, she will deposit between \$30 and \$60 into the account. Which inequality represents all possibilities for the total amount of money x , in dollars, in Imani's account after 10 months of deposits this year?

- A) $x \leq 300$
- B) $x \geq 600$
- C) $170 \leq x \leq 620$
- D) $470 \leq x \leq 770$

Question 14

The function f is defined by $f(x) = (x - 2)(x - 8)(x + 3)$. In the xy -plane, the graph of $y = h(x)$ is the result of translating the graph of $y = f(x)$ up 6 units. What is the y -coordinate of the y -intercept of the graph of $y = h(x)$?

- A) 54
- B) 48
- C) 6
- D) 0

Question 15

A circle has a circumference of 28 centimeters. What is the diameter, in centimeters, of the circle, rounded to the nearest hundredth?

Question 16

How many solutions does the equation $35x - 25 = 5(7x - 5)$ have?

- A) Exactly one
- B) Exactly two
- C) Infinitely many
- D) Zero

Question 17

On January 1, 2000, the population of a town was 26,257 and on January 1, 2010, the population was 26,858. The equation $10x + 26,257 = 26,858$ describes this situation. Which of the following is the best interpretation of x in this context?

- A) The total increase in population between 2000 and 2010
- B) The projected population of the town 10 years after 2010
- C) The average increase per year of the population between 2000 and 2010
- D) The percentage by which the population of the town increased each year between 2000 and 2010

Question 18

For the linear function f , $f(-5) = -7$, and the slope of the graph of $y = f(x)$ in the xy -plane is 3. Which equation defines f ?

- A) $f(x) = 3x + 8$
- B) $f(x) = 3x - 5$
- C) $f(x) = 3x - 7$
- D) $f(x) = 3x - 12$

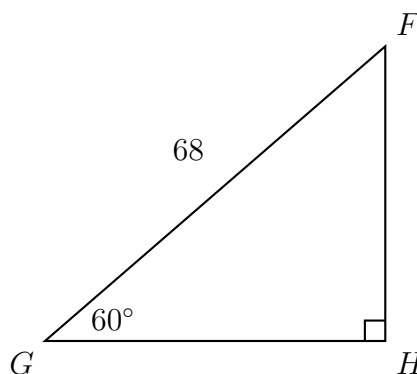
Question 19

In the xy -plane, an equation of circle A is $(x - 2)^2 + (y - 7)^2 = 25$. Circle B has the same center as circle A but has a radius that is twice the radius of circle A . Which equation represents circle B ?

- A) $(x - 2)^2 + (y - 7)^2 = 50$
- B) $(x - 2)^2 + (y - 7)^2 = 100$
- C) $(x - 2)^2 + (y - 7)^2 = 250$
- D) $(x - 2)^2 + (y - 7)^2 = 625$

Question 20

Line k contains the points $(-3, -60)$, $(v, 0)$, and $(5, 76)$. What is the value of v ?

Question 21

Note: Figure not drawn to scale.

In right triangle FGH shown, $\angle G = 60^\circ$ and hypotenuse $FG = 68$. What is the value of $\cos F$?

- A) $\frac{\sqrt{3}}{68}$
- B) $\frac{1}{2}$
- C) $\frac{\sqrt{3}}{2}$
- D) $34\sqrt{3}$

Question 22

Based on a model, on day 1 of an 8-day experiment, a lima bean plant had an estimated mass of 253 grams, and each day after day 1 of the experiment, the estimated mass of the plant decreased by 4 grams. Which equation represents this model, where m is the estimated mass, in grams, of the plant on day x of the experiment and $1 \leq x \leq 8$?

- A) $m = -4x + 245$
- B) $m = -4x + 249$
- C) $m = -4x + 253$
- D) $m = -4x + 257$

Module 2

Question 1

A cable company charges new customers a one-time installation fee and a monthly service fee for cable service. The equation $y = 50x + 120$ gives the total amount y , in dollars, the cable company charges new customers for x months of cable service. What is the amount, in dollars, of the one-time installation fee the cable company charges new customers?

- A) 170
- B) 120
- C) 70
- D) 0

Question 2

Data set X: 13, 16, 19, 21, 24, 25, 25, 26, 38

Data set Y: 13, 16, 19, 21, 24, 25, 25, 26, 33

Data set Y is created by replacing the number 38 in data set X with the number 33. Which of the following statements is true about the means and medians of data set X and data set Y?

- A) The mean of data set X is greater than the mean of data set Y, and the median of data set X equals the median of data set Y.
- B) The mean of data set X is greater than the mean of data set Y, and the median of data set X is greater than the median of data set Y.
- C) The mean of data set X equals the mean of data set Y, and the median of data set X equals the median of data set Y.
- D) The mean of data set X is less than the mean of data set Y, and the median of data set X is greater than the median of data set Y.

Question 3

$$y = 4x + 36$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which equation could be the second equation in this system?

- A) $y - 4x = -36$
- B) $y - 4x = 36$
- C) $y - 9x = 18$
- D) $y - 9x = 0$

Question 4

If $x + 3y = 29$ and $7x - 12y = -61$, what is the value of y ?

Question 5

$$R = \frac{0.36L}{A}$$

The given equation relates the resistance R , in ohms, of a wire to its length l , in meters, and its cross-sectional area A , in square meters. Which equation correctly expresses the length, in meters, of the wire in terms of its resistance, in ohms, and its cross-sectional area, in square meters?

- A) $L = (R - A) / 0.36$
- B) $L = (R + A) / 0.36$
- C) $L = 0.36R / A$
- D) $L = AR / 0.36$

Question 6

For the polynomial function f , the graph of $y = f(x)$ in the xy -plane passes through the points $(-5, 0)$, $(3, 0)$, and $(7, 0)$. Which of the following must be a factor of $f(x)$?

- A) $x + 3$
- B) $x + 7$
- C) $x - 3$
- D) $x - 5$

Question 7

A line in the xy -plane passes through the origin and the points $(k, 8)$ and $(18, k)$, where $k > 0$. What is the value of k ?

Question 8

In the xy -plane, line k has a slope of $\frac{6}{13}$, an x -intercept of $(-6, 0)$, and a y -intercept of $(0, p)$. What is the value of p ?

Question 9

The measure of angle S is $\frac{8\pi}{11}$ radians. The measure of angle T is 2 times the measure of angle S . Which expression represents the measure, in degrees, of angle T ?

- A) $\frac{8}{11} \cdot 90^\circ \cdot 2$
- B) $\frac{8}{11} \cdot 180^\circ \cdot 2$
- C) $\frac{8\pi}{11} \cdot 90^\circ \cdot 2$
- D) $\frac{8\pi}{11} \cdot 180^\circ \cdot 2$

Question 10

What is the x -intercept of the graph of $y = \frac{45 - x}{ax + b}$ in the xy -plane, where a and b are positive constants?

- A) $\left(-\frac{b}{a}, 0\right)$
- B) $\left(0, \frac{45}{b}\right)$
- C) $(0, -45)$
- D) $(45, 0)$

Question 11

The ratio x to y is equivalent to the ratio 4 to 7. If $x = 11t$, what is the value of y in terms of t ?

- A) $\frac{4}{77}t$
- B) $\frac{7}{44}t$
- C) $\frac{44}{7}t$
- D) $\frac{77}{4}t$

Question 12

A research manager selected 2 random samples of ovens of a certain type to estimate the average amount of time this type of oven takes to preheat to 350 degrees Fahrenheit (°F). Based on the first sample, the research manager estimated that this type of oven takes an average of 14.2 minutes to preheat to 350°F, with an associated margin of error of 1 minute. Based on the second sample, the research manager estimated that this type of oven takes an average of 14.4 minutes to preheat to 350°F, with an associated margin of error of 2.2 minutes. Assuming the margins of error were calculated the same way, which of the following best explains why the first sample obtained a smaller margin of error than the second sample?

- A) The first sample contained fewer ovens than the second sample.
- B) The first sample contained more ovens than the second sample.
- C) The first sample took less time on average to preheat to 350°F than the second sample.
- D) The first sample took more time on average to preheat to 350°F than the second sample.

Question 13

$$f(x) = 23(1.20)^{x/4}$$

For the given function f , the value of $f(x)$ increases by $p\%$ for every increase of x by 8. What is the value of p ?

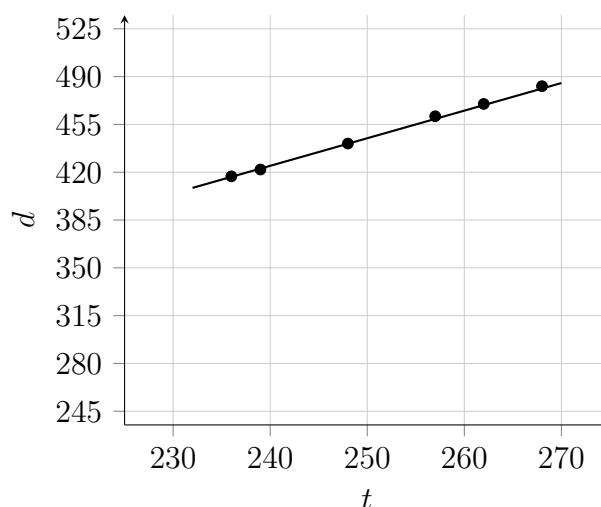
- A) 20
- B) 31
- C) 40
- D) 44

Question 14

A model estimates that in a particular forest, the number of trees with any given diameter measured at shoulder height is 21% less for each 1-inch increase in tree diameter measured at shoulder height. The model can be written in the form $f(x) = ab^x$, where a and b are constants and x is the tree's diameter, in inches, measured at shoulder height, and $x \geq 5$. The model estimates that 3,100 trees in this forest have a diameter of 13 inches measured at shoulder height. Which function best represents this model?

- A) $f(x) = 3,100 \cdot 0.21^x$
- B) $f(x) = 3,100 \cdot 0.79^x$
- C) $f(x) = 66,000 \cdot 0.21^x$
- D) $f(x) = 66,000 \cdot 0.79^x$

Question 15



The scatterplot shows the relationship between two variables, t and d . Which of the following equations is the most appropriate linear model for the data shown?

- A) $d = -60.1 + 2.02t$
- B) $d = 160.1 + 2.02t$
- C) $d = 359.8 + 2.02t$
- D) $d = 394.8 + 2.02t$

Question 16

The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ passes through the points $(c, 11)$ and $(2c, 221)$, where c is a constant. Which of the following could be the value of b ?

- A) 4
- B) 25
- C) 210
- D) 232

Question 17

The price of an item increased by $p\%$ from \$90 to \$93. What is the value of p ?

Question 18

In triangle ABC , the measure of angle A is 56° and $AC = 30$. In triangle PQR , the measure of angle P is 56° and $PR = 90$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

- A) $AB = 20$ and $PQ = 20$.
- B) $AB = 20$ and $QR = 60$.
- C) The measures of angle B and angle R are 48° and 76° , respectively.
- D) The measures of angle B and angle Q are 56° and 48° , respectively.

Question 19

$$\frac{5}{9}(x+9)(x+\sqrt{5k+9})(x-\sqrt{5k+9})=0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 81. What is the value of k ?

Question 20

$$4x^2 - px + w = -86$$

In the given equation, p and w are integer constants. The equation has exactly one real solution. Which is **NOT** a possible value of w ?

- A) -22
- B) 14
- C) 25
- D) 314

Question 21

If $\frac{x+6}{5} = \frac{x+6}{13}$, the value of $x+6$ is between which of the following pairs of values?

- A) -7 and -5
- B) -2 and 2
- C) 2 and 7
- D) 8 and 13

Question 22

The expression $6x^4 + 17x^2 + 5$ can be rewritten as $(3x^2 + a)(x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(x^2 + d)$, where c and d are positive non-integers. What is the value of $a + c$?

Answer Sheet

Module 1

1	2	3	4	5	6	7	8	9	10
<i>C</i>	<i>A</i>	<i>B</i>	<i>D</i>	3800	1.2	<i>C</i>	<i>A</i>	<i>D</i>	<i>A</i>

11	12	13	14	15	16	17	18	19	20
43	343	<i>D</i>	<i>A</i>	8.91	<i>C</i>	<i>C</i>	<i>A</i>	<i>B</i>	0.529

21	22
<i>C</i>	<i>D</i>

Module 2

1	2	3	4	5	6	7	8	9	10
<i>B</i>	<i>A</i>	<i>B</i>	8	<i>D</i>	<i>C</i>	12	36/13	<i>B</i>	<i>D</i>

11	12	13	14	15	16	17	18	19	20
<i>D</i>	<i>B</i>	<i>D</i>	<i>D</i>	<i>A</i>	<i>A</i>	10/3	<i>C</i>	36/5	<i>C</i>

21	22
<i>B</i>	17/2